

Decode lab IOT

Project#04:

Edge-Computing Smart Home Appliance with Interrupts and Safety Override

Objective:

To build a smart edge-computing system that:

- Detects human motion instantly
- Controls lighting automatically
- Detects gas/smoke leakage
- Activates emergency safety mode
- Uses hardware interrupts for fast response

Components Required:

Component

Quantity

Arduino UNO

1

PIR Motion Sensor

1

MQ-2 Gas Sensor

1

White LED

1

Red LED

1

Buzzer

1

220Ω Resistors

2

Breadboard

1

Jumper Wires

Multiple

Circuit Connections:

PIR Sensor Connections

PIR Pin

Arduino

VCC

5V

GND

GND

OUT

D2

MQ-2 Gas Sensor Connections

MQ-2 Pin

Arduino

VCC

5V

GND

GND

AO

A0

LEDs

LED

Arduino Pin

White LED

D8

Red LED

D9

Buzzer

Buzzer

Arduino

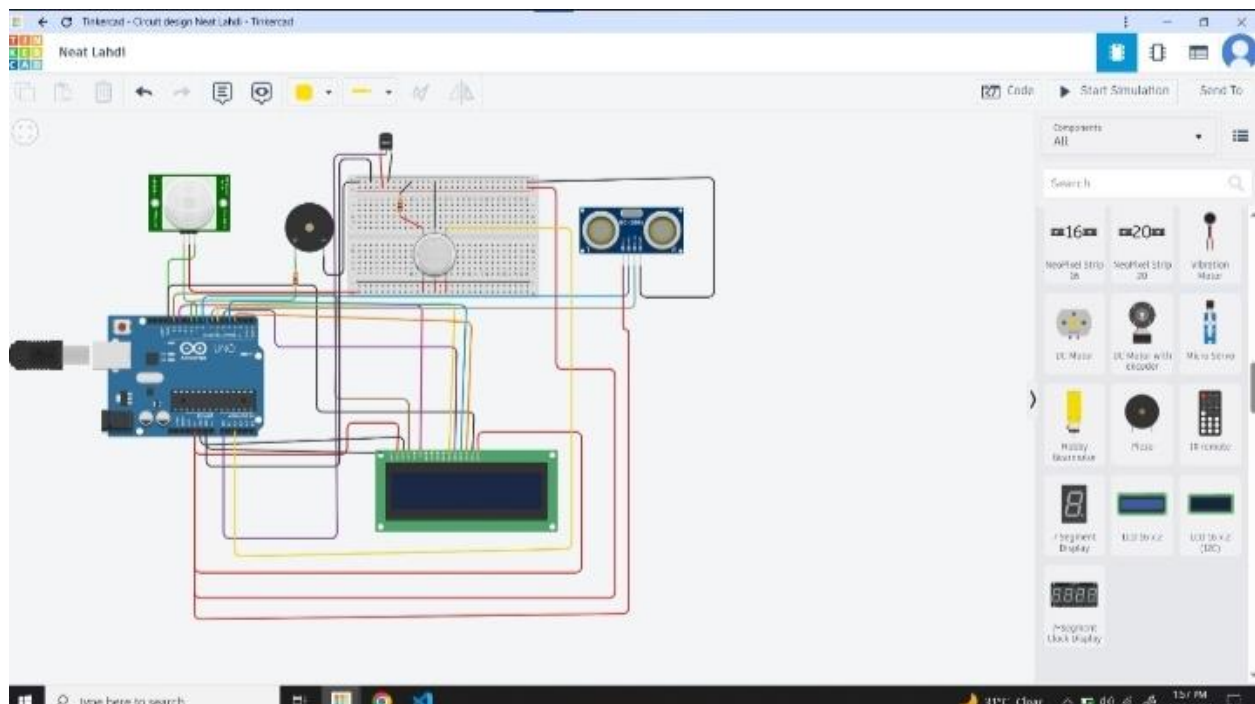
Positive

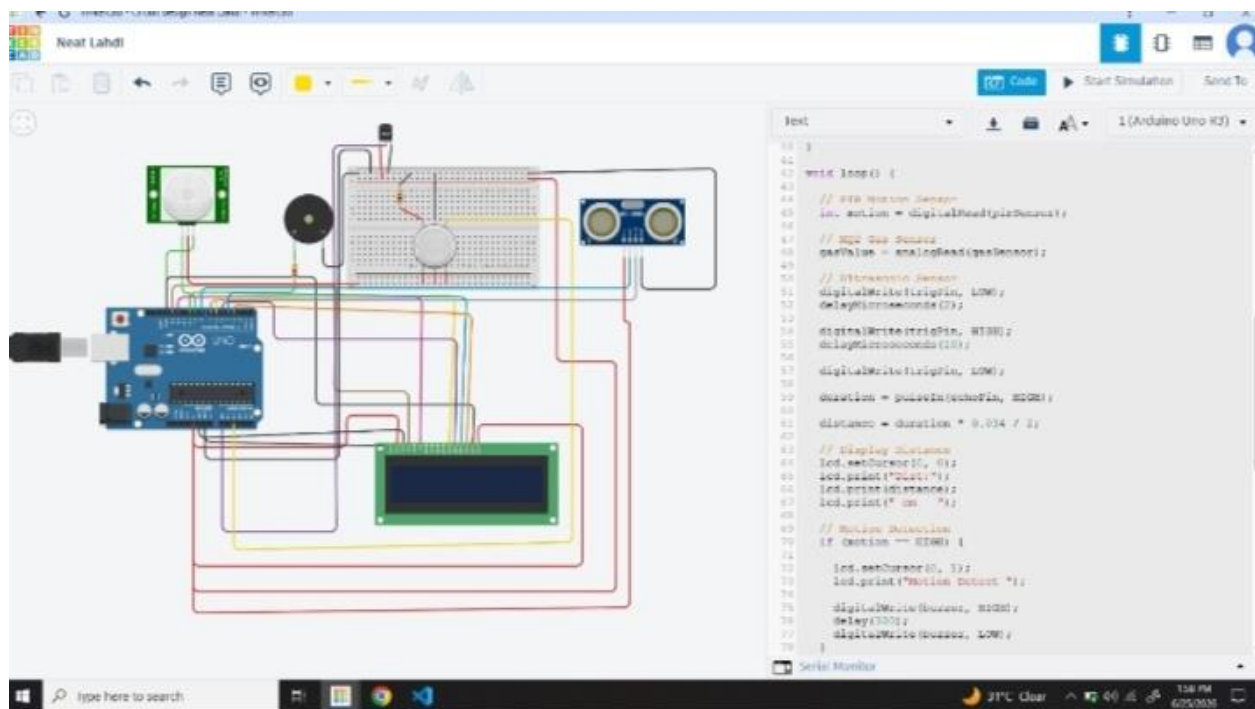
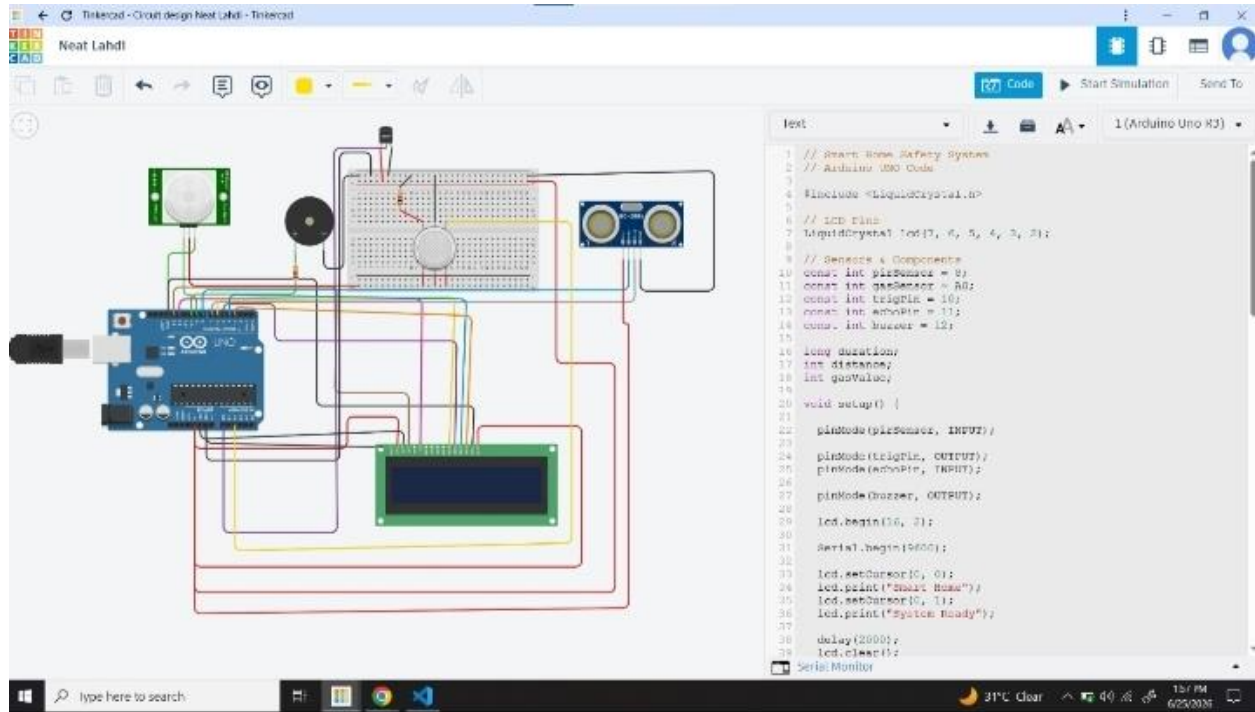
D10

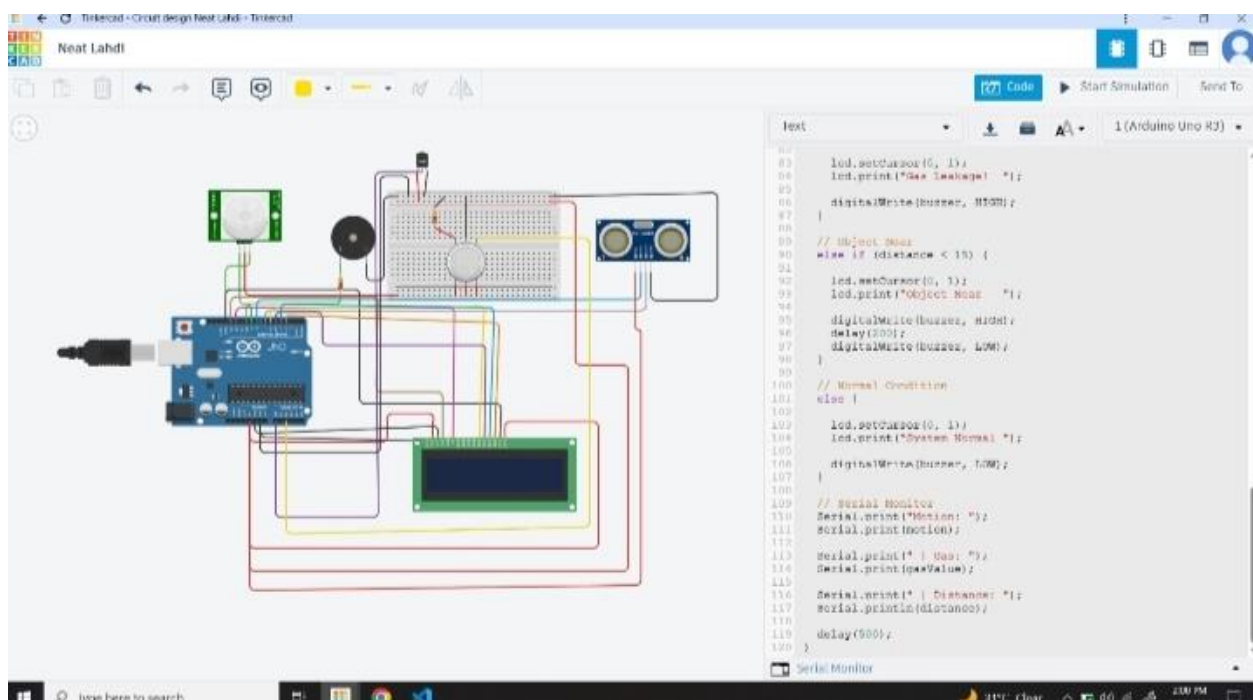
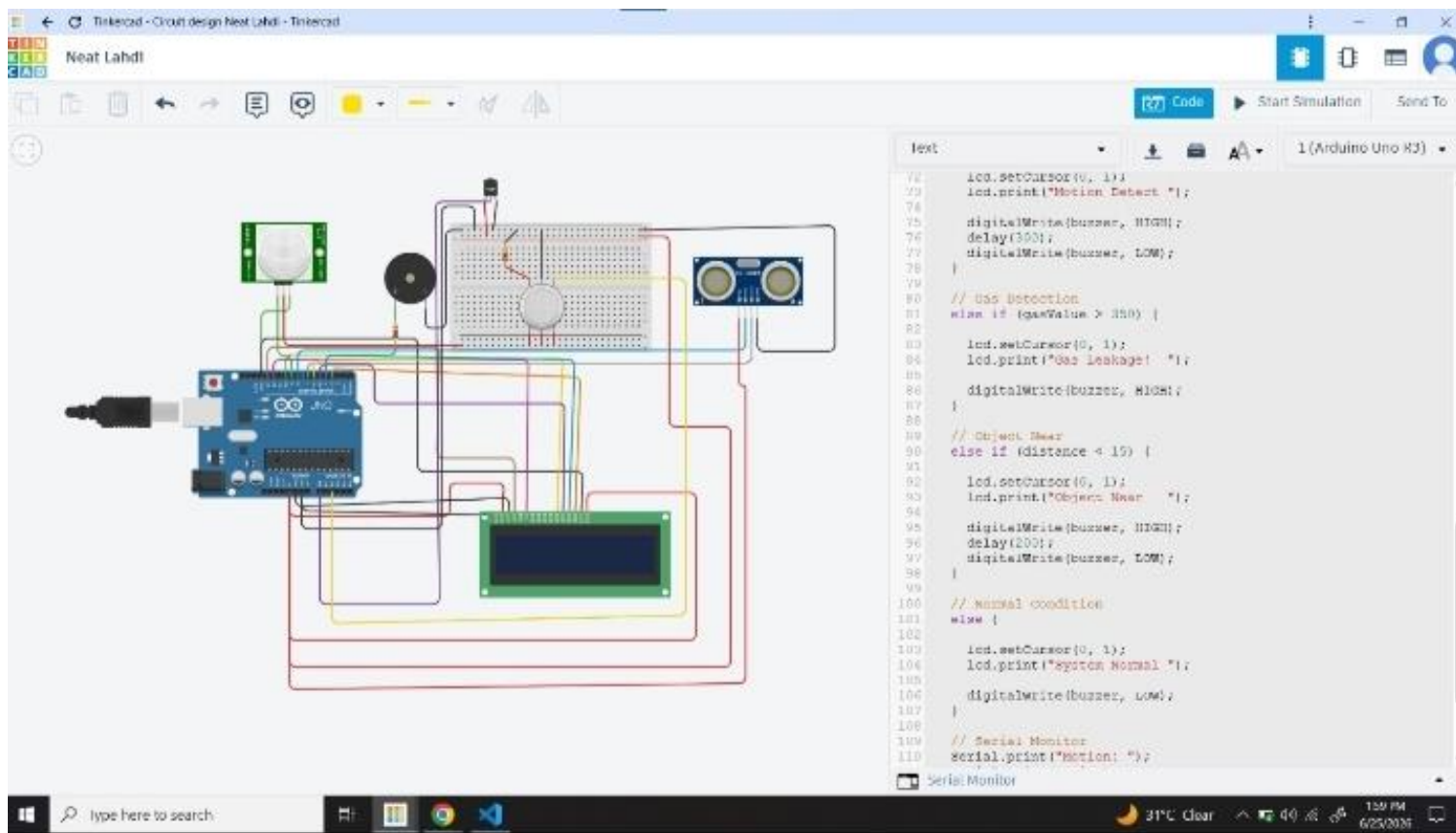
Negative

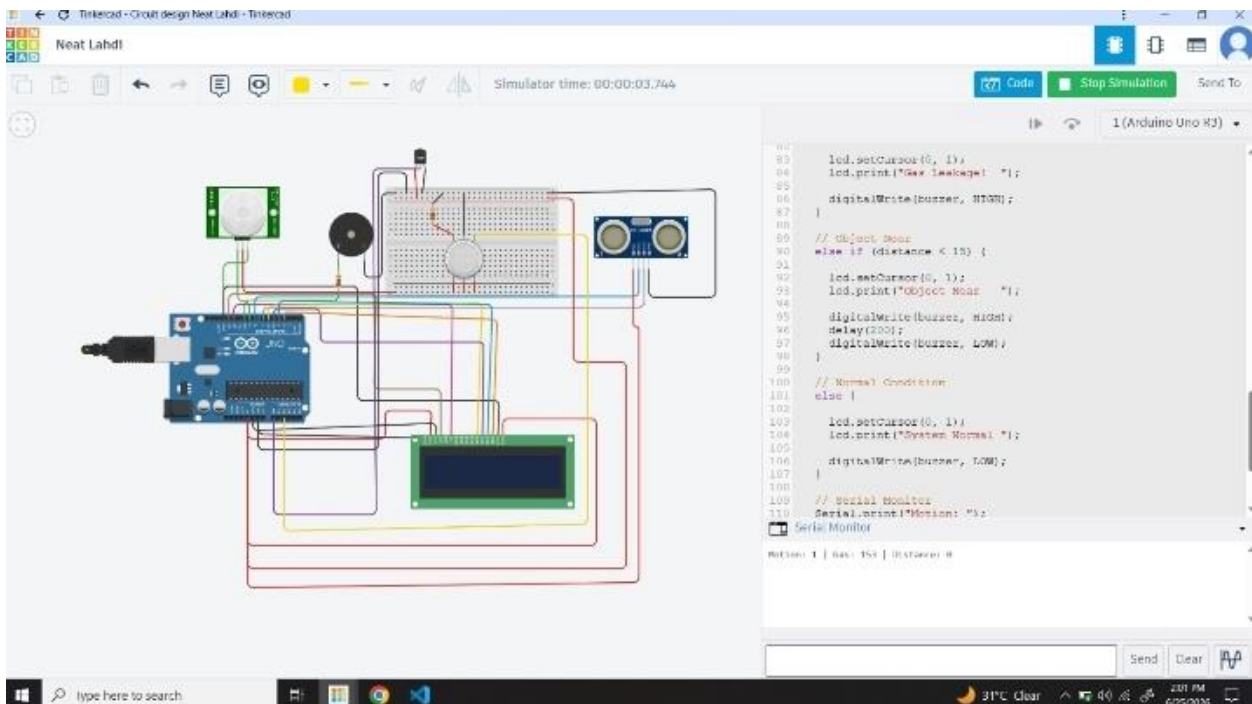
GND

Circuit Diagram:










```
// LCD Pins
```

```
LiquidCrystal lcd(7, 6, 5, 4, 3, 2);
```

```
// Sensors & Components
```

```
Const int pirSensor = 8;
```

```
Const int gasSensor = A0;
```

```
Const int trigPin = 10;
```

```
Const int echoPin = 11;
```

```
Const int buzzer = 12;
```

```
Long duration;
```

```
Int distance;
```

```
Int gasValue;
```

```
Void setup() {
```

```
    pinMode(pirSensor, INPUT);
```

```
    pinMode(trigPin, OUTPUT);
```

```
    pinMode(echoPin, INPUT);
```

```
    pinMode(buzzer, OUTPUT);
```

```
    lcd.begin(16, 2);
```

```
    Serial.begin(9600);
```

```
    Lcd.setCursor(0, 0);
```

```
    Lcd.print("Smart Home");
```

```
Lcd.setCursor(0, 1);
```

```
Lcd.print("System Ready");
```

```
Delay(2000);
```

```
Lcd.clear();
```

```
}
```

```
Void loop() {
```

```
  // PIR Motion Sensor
```

```
  Int motion = digitalRead(pirSensor);
```

```
  // MQ2 Gas Sensor
```

```
  gasValue = analogRead(gasSensor);
```

```
// Ultrasonic Sensor
```

```
digitalWrite(trigPin, LOW);
```

```
delayMicroseconds(2);
```

```
digitalWrite(trigPin, HIGH);
```

```
delayMicroseconds(10);
```

```
digitalWrite(trigPin, LOW);
```

```
duration = pulseIn(echoPin, HIGH);
```

```
distance = duration * 0.034 / 2;
```

```
// Display Distance
```

```
Lcd.setCursor(0, 0);
```

```
Lcd.print("Dist:");
```

```
Lcd.print(distance);
```

```
Lcd.print(" cm ");
```

```
// Motion Detection
```

```
If (motion == HIGH) {
```

```
    Lcd.setCursor(0, 1);
```

```
    Lcd.print("Motion Detect ");
```

```
    digitalWrite(buzzer, HIGH);
```

```
    delay(300);
```

```
    digitalWrite(buzzer, LOW);
```

```
}
```

```
// Gas Detection
```

```
Else if (gasValue > 350) {
```

```
    Lcd.setCursor(0, 1);
```

```
    Lcd.print("Gas Leakage! ");
```

```
    digitalWrite(buzzer, HIGH);
```

```
}
```

```
// Object Near
```

```
Else if (distance < 15) {
```

```
    Lcd.setCursor(0, 1);
```

```
    Lcd.print("Object Near ");
```

```
digitalWrite(buzzer, HIGH);  
delay(200);  
digitalWrite(buzzer, LOW);  
}
```

```
// Normal Condition
```

```
Else {
```

```
Lcd.setCursor(0, 1);
```

```
Lcd.print("System Normal ");
```

```
digitalWrite(buzzer, LOW);  
}
```

```
// Serial Monitor
```

```
Serial.print("Motion: ");
```

```
Serial.print(motion);
```

```
Serial.print(" | Gas: ");
```

```
Serial.print(gasValue);
```

```
Serial.print(" | Distance: ");
```

```
Serial.println(distance);
```

```
Delay(500);
```

```
}
```

Flowchart:

START

|

V

Initialize Arduino System

|

V

Read MQ-2 Gas Sensor

|

±-----┴-----┴



Gas Detected?

No Gas



YES

v

|

Motion Detected?



V

±-----┴-----┴

Activate Buzzer

|

|

Flash Red LED

YES

NO

Turn OFF White LED

|

|

Emergency Mode v |

Turn ON White LED |

Wait 5 Seconds |

Turn OFF White LED |



±----→ Repeat

Working Explanation:

Normal Mode

- System continuously checks PIR and MQ-2 sensors.

- White LED remains OFF.

Motion Detection

When a person enters:

- PIR sensor sends interrupt signal.
- ISR executes immediately.

- White LED turns ON for 5 seconds.

Safety Override Mode

If gas/smoke is detected:

- System ignores motion logic.
- White LED turns OFF.
- Red LED flashes.
- Buzzer sounds continuously.

This creates:

- Industrial fail-safe behavior
- Real-time emergency response

Output:

Motion Detection Output

Serial Monitor

System Started

Gas Sensor Value: 120

Motion Detected

Hardware Output

- White LED ON
- Red LED OFF
- Buzzer OFF

Gas Detection Output

Serial Monitor

Gas Sensor Value: 480

WARNING! GAS LEAK DETECTED

Hardware Output

- White LED OFF
- Red LED FLASHING
- Buzzer ON

Edge Computing Concept:

This project uses edge computing because:

- Decisions happen locally on Arduino
- No internet/cloud required
- Faster response time
- Better safety performance

Industrial Fail-Safe Logic:

Gas sensor has highest priority.

If danger occurs:

- System overrides all other operations
- Alarm activates immediately
- Lighting automation stops

This is called:

- Safety Override
- Fail-Safe Architecture

Advantages:

Feature

Benefit

Interrupts

Instant response

Gas Detection

Safety improvement

Smart Lighting

Energy saving

Edge Processing

Fast operation

Multi-Sensor Logic

Better automation

Applications:

- Smart homes
- Factories
- Warehouses

- Smart offices
- Industrial safety systems

Conclusion:

This project successfully demonstrates:

- Hardware interrupts
- Smart automation
- Edge computing
- Safety override systems
- Real-time embedded processing